

Midterm Project 1

This is an *individual* effort. You can come to me or the TAs for help if you get stuck, but you should not be working in groups. Class notes and anything linked from the class ELMS page can be used as a resource. Submit both your RScript code and the requested word document with your name and the assignment in the file name.

Section 1: Binomial Data

Description:

Cancer in dogs: A study in 1994 examined 491 dogs that had developed cancer and 945 dogs as a control group to determine whether there is an increased risk of cancer in dogs that are exposed to the herbicide 2,4-Dichlorophenoxyacetic acid (2,4-D). Order: a factor with levels 2,4-D and no 2,4-D, response: a factor with levels cancer and no cancer. This was an observational study.

[Dataset](#)

[Download Dataset](#)

Questions: Write your answers in a word file to be submitted with your RScript file.

1. Identify the control group and the treatment group.
2. Write a hypothesis for what you think the effect of exposure to the herbicide might be as shown in class both in words and mathematically.
3. Now test that hypothesis. Select an alpha, choose the correct binomial hypothesis test and run it.
4. What are your results? Express these as shown in class.
5. Does this study make it possible to prove causation? Why or why not?

Section 2: Test of Means

This data approaches student achievement in secondary education of two Portuguese schools. The data attributes include student grades, demographic, social and school related features, and it was collected by using school reports and questionnaires. Two datasets are provided for two distinct subjects: Mathematics and Portuguese.

Datasets: [Portuguese](#)

[Download Portuguese](#)

, [Mathematics](#)

[Download Mathematics](#)

[Data Dictionary](#)

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1. Compare the mean G3 grades (the equivalent of the semester grade) for Mathematics and Portuguese.
 1. What test are you going to run?
 2. Have you met all assumptions?
 3. What are your null and research hypotheses and your alpha in words and in mathematical notation?
 4. What are your results?
2. Compare the G1 and G2 mean grades for ***either*** Mathematics or for Portuguese (just one of the datasets). Each row of data represents information for a single student.
 1. What test are you going to run?
 2. Have you met all assumptions?
 3. What are your null and research hypotheses and your alpha in words and in mathematical notation?
 4. What are your results?

Section 3: Concepts

1. Which of the following IS/ARE NOT a measure of central tendency?
 1. Sample
 2. Mean
 3. Range
 4. Median
 5. Mode
2. Define each of the following:
 1. Probability distribution
 2. Population
 3. Sample
 4. Parameter
 5. Variable
3. For each of the following, identify the variable type:
 1. Dice rolls
 2. Lyme disease infection rate
 3. Satisfaction survey rankings
 4. Whether or not a dog got parvo
 5. IQ scores
 6. Ethnicity
 7. Height

8. Income

4. Describe three types of probability sampling and when you would use each.
5. If a test is listed as robust with respect to normality, what does that mean?
6. Identify three ways in which statistics can be used incorrectly or misleadingly. Give an example of each. How would you respond if asked to do one of these things at a job?
7. Why is a large sample size an advantage? What problems does it solve? Why do researchers not always take large samples?
8. Under what (if any) circumstances can causation be proved?
9. I have two normally distributed samples from two different populations. I do not know if the populations have the same variance. What test would I use to compare the means of those samples and why?
10. What happens if you violate the assumptions of a test?
11. Can a confidence interval take the place of a t-test? Why or why not?